# Temporal Data Lineage and Impact Analysis in Oracle Database: Evolution to Oracle Graph Architecture

Rovinj | 2026-10-15



Part 1: Operational (SQL) Property Graphs inside the Oracle AI Database


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Part 2: Temporal Data Lineage and Impact Analysis using Visual Interactive Analysis


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# Part 1: The Oracle AI Database as a Graph Database - Defining and Analyzing Operational (SQL) Property Graphs


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# ORACLE 26ai AI Database

## Data Types

Vector · Spatial · JSON · Text · XML · Relational

## Workloads

AI / ML · IoT · Analytical · Streaming · Graph · Blockchain · Distributed

## Tools

APEX · ORDS · SQLcl · MCP Server · AutoML · Enterprise Manager · SQL Developer VS Code · Agent Factory · Private Contexts

## Dev Interfaces

SQL First · MongoDB API · Language Support

## Runs Anywhere

On-Premises · Containers · Oracle Cloud · Autonomous AI Database · Cloud@Customer · @Azure / @AWS / @GCP

05
**Oracle Graphs + APEX**
Property graphs, interactive explorer.
+ Graph analytics
+ Health dashboard
+ AI explanations

04
**Output Tables**
Lineage · Impact · Intersections.

02
**SCD2 Tracking**
Track changes, time-travel.

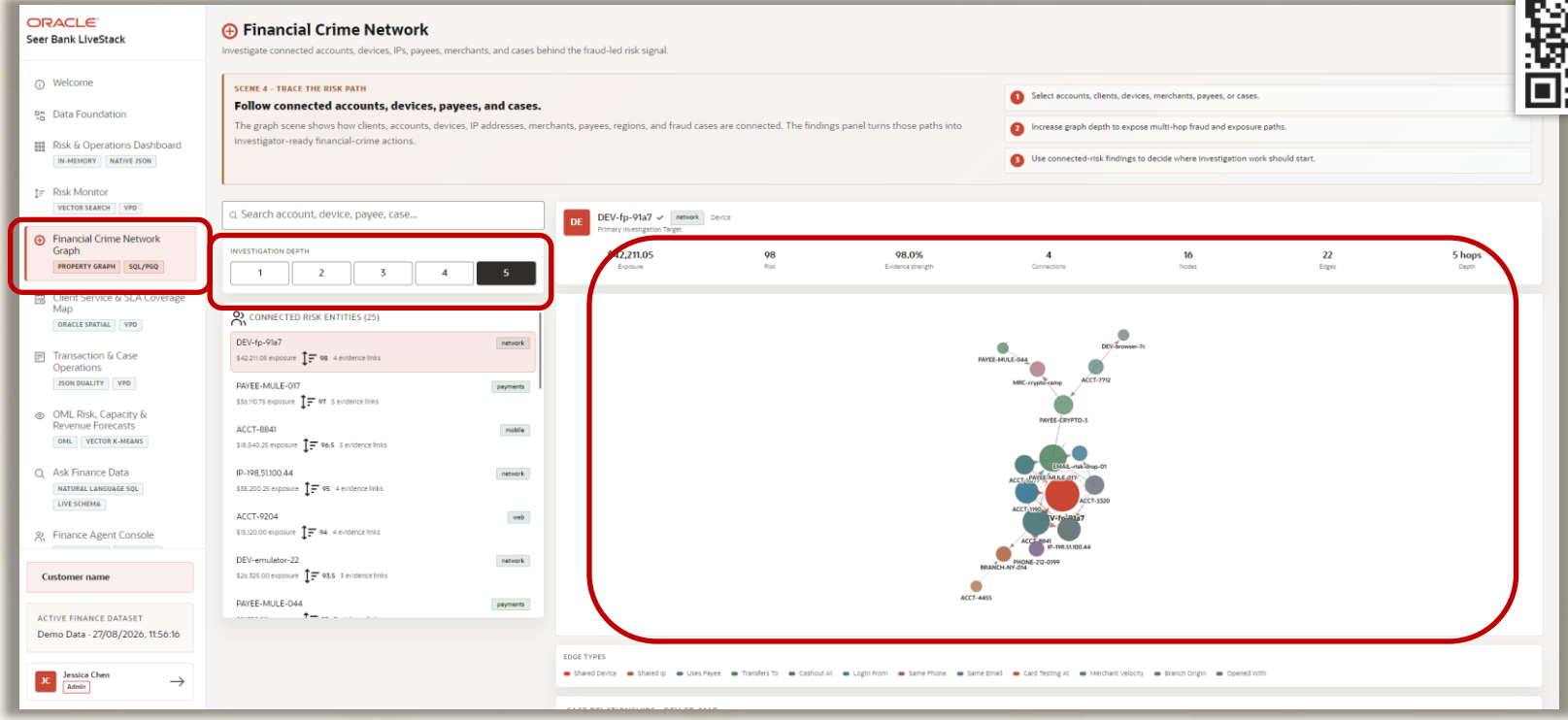
03
**Graph Inference**
Build nodes/edges, run inference.

05
**Oracle Graphs + APEX**
Property graphs, interactive explorer.
+ Graph analytics
+ Health dashboard
+ AI explanations


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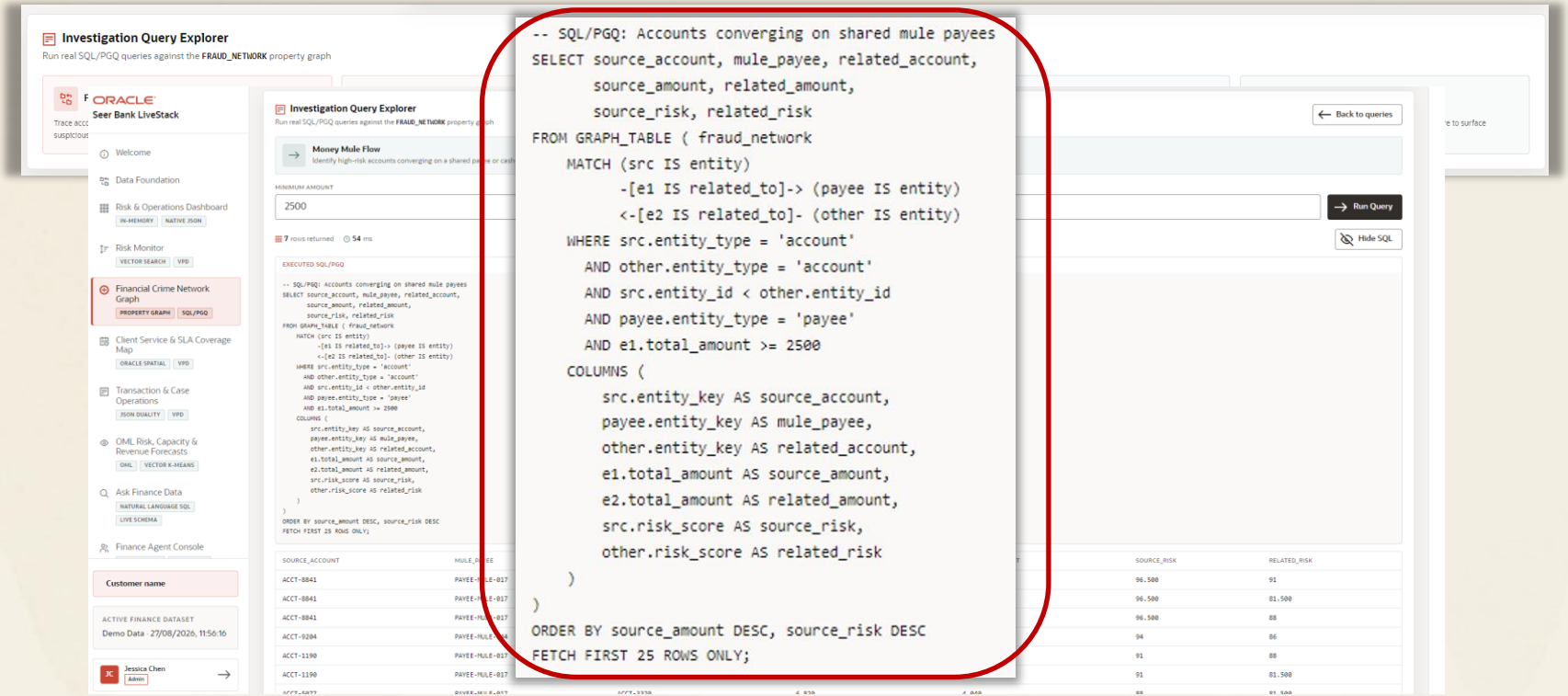
# Converged Database in Action ☞LiveLabs Workshop

**Build Financial Crime and Operations Intelligence on One Governed Oracle Data Platform**



# Converged Database in Action ☞LiveLabs Workshop

## Build Financial Crime and Operations Intelligence on One Governed Oracle Data Platform



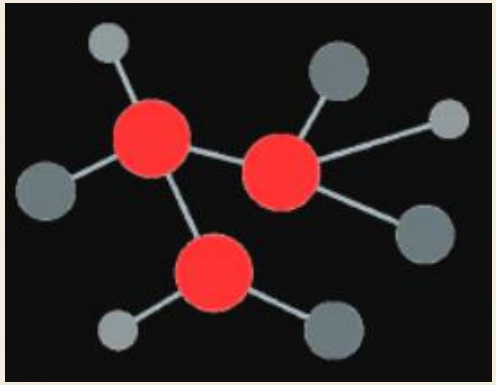
# Fundamentals




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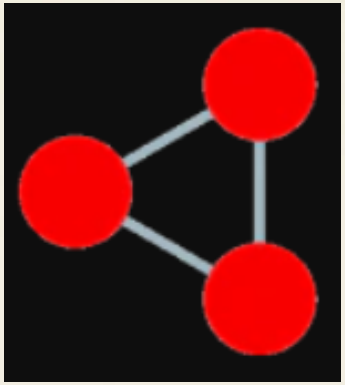
# Graph Support in the Oracle Database

No additional license costs
All editions of the Oracle DB
All Oracle DB Cloud Services



## (Labeled) Property Graphs

- Since Oracle Database version 11.2
- Generic graph model
- Keywords: Graph Pattern-Matching, Graph Algorithms, Graph Machine Learning, PGQL, SQL/PGQ

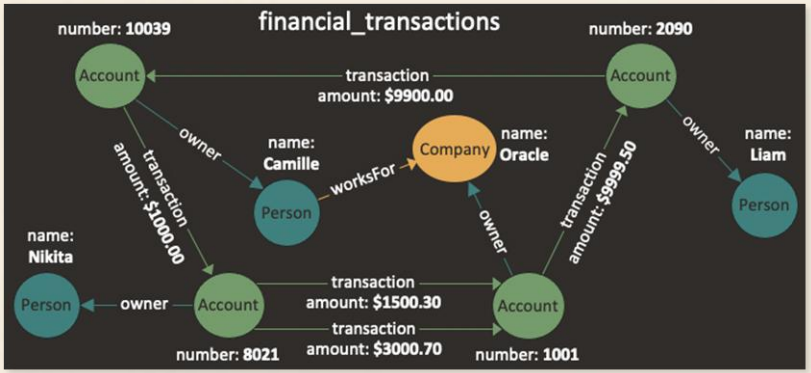


## RDF Knowledge Graphs

- Since Oracle Database version 10.2
- Specialized, standards-based graph model
- Keywords: Semantic technologies, Linked Data, SPARQL, RDF, RDF-S, OWL, Ontologies, Inferencing


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# The Property Graph Data Model




Source: pgql-lang.org

A representation of the real world, focusing on relationships using

- Vertices (nodes)
  representing a *Thing*
- Edges
  representing connections between vertices
- Properties
  expressed as key-value pairs attached to vertices and/or edges
- Labels
  representing the type of a vertex or edge


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# Create and Analyze Property Graphs using SQL




## Prerequisite: Oracle Database 23ai or higher

- Operational (SQL) Property Graphs using SQL/PGQ
- SQL/PGQ is an extension to SQL:2023 (ISO/IEC 9075-16)

## Step 1: Create a graph

CREATE PROPERTY GRAPH to build graphs as **metadata objects** over the original data

- Work with data where it lives ☞ No data copy or move

## Step 2: Analyze the graph

MATCH clause inside the GRAPH_TABLE function to run Pattern-Matching Queries

Latest addition: DBMS_OGA to run **Graph Algorithms** (PageRank and variants, Bellman-Ford, Weakly Connected Components)

## Schema flexibility through JSON

- JSON document as a property
- JSON element as a property

## Combines all converged DB features using SQL

See also:
www.oracle.com/a/ocom/docs/database/operational_property_graph_with_23c.pdf
blogs.oracle.com/database/post/property-graphs-in-oracle-database-26ai-the-sql-pgq-standard


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# Tools




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# Tools to work with SQL Property Graphs in 26ai

All your existing **SQL tools** work with graphs

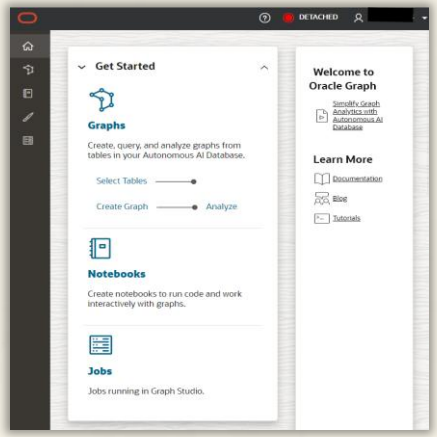
- SQL Developer (**Latest: 26.2**), SQLcl
- VSCode
  - SQL Developer Extension
  - SQL Developer Graph Visualization Extension
- Graph Visualization JavaScript Library
- APEX (Plug-in, Sample Graph Viz app)
- Advanced queries, graph algorithms

**Graph Server and Clients (Latest: 26.3)**
providing additional capabilities

- APIs and CLIs
- Graph Algorithms
- Graph Machine Learning
- Graph Visualization Web App (*User-managed Graph Studio*)
- Graph Visualization JavaScript Library
- Integration with Jupyter Notebook/Lab

## Graph Studio

- Autonomous AI Database integrated tool ☞ *Graph Studio*
  - Automated and AI-supported graph modeling
  - Built-in notebook support, including visualization tools




oracle / skills

github.com/oracle/skills > db > appdev > sql-property-graph.md


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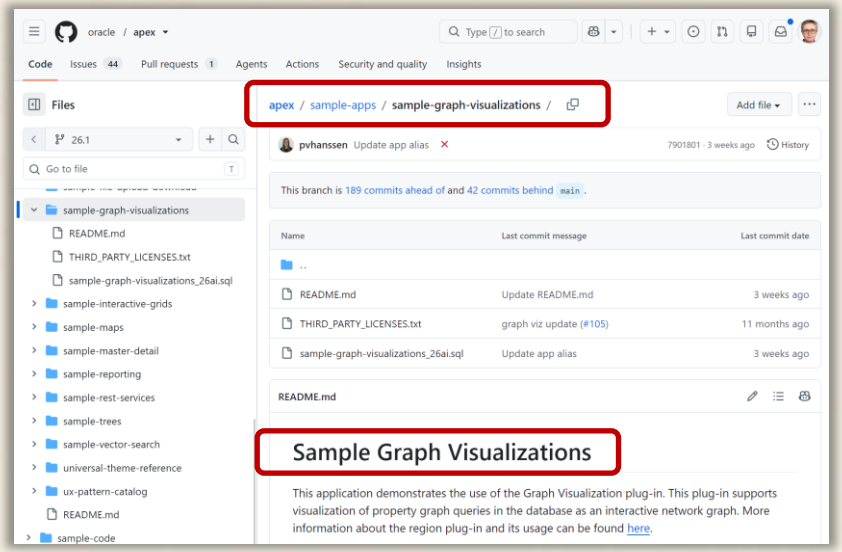
# Graph-Enabled Low-Code App Dev

**Property Graphs natively supported in Oracle Application Express (APEX)**

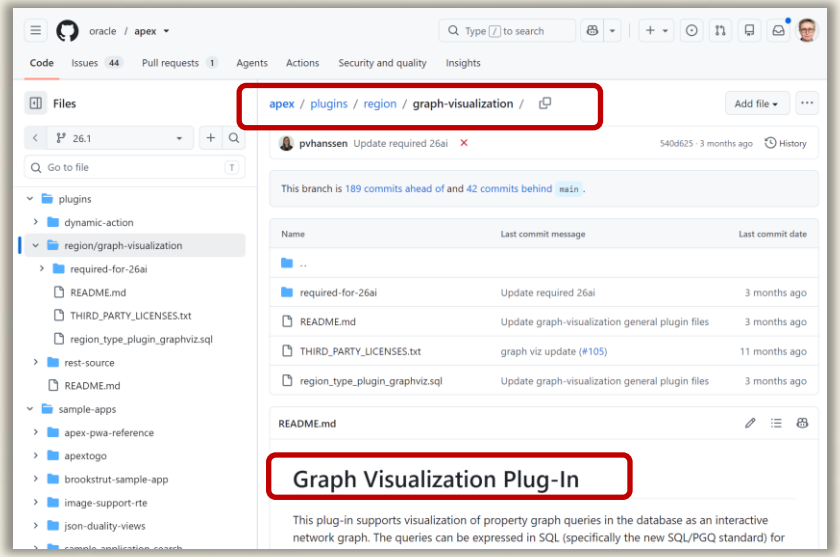



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# Sample Graph Visualization App and Plugin



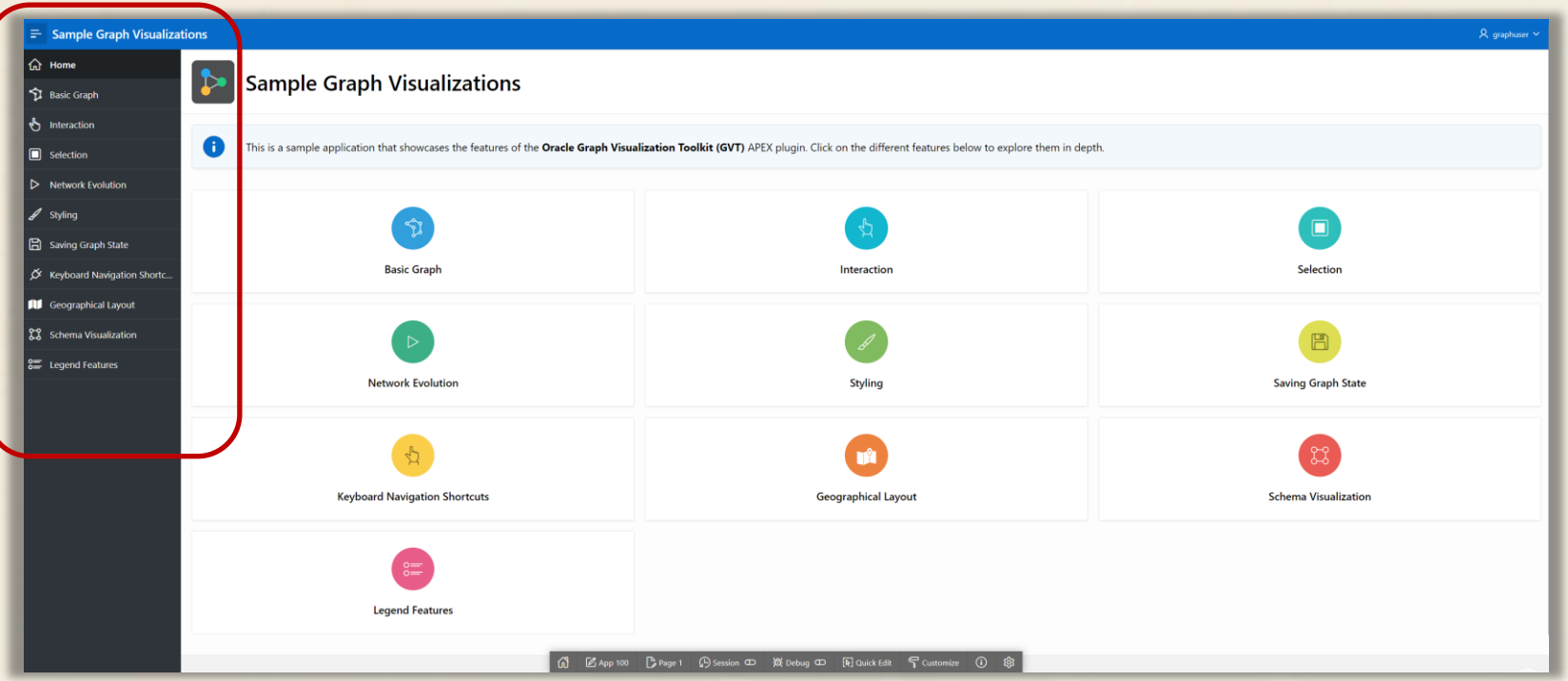

github.com/oracle/apex/tree/26.1/sample-apps/sample-graph-visualizations




github.com/oracle/apex/tree/26.1/plugins/region/graph-visualization


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# Sample Graph Viz app – Home





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# Sample Graph Viz app – Schema Visualization





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# KOIOS VIA Property Graphs

## The LINEAGE Graph under the Hood


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# Create the LINEAGE Graph

```
CREATE PROPERTY GRAPH via_lineage_graph
VERTEX TABLES (
  mv_via_nodes_columns AS column_node
    KEY (node_id)
    LABEL column_node
    PROPERTIES (schema_name, table_name, column_name, full_name)
)
EDGE TABLES (
  mv_via_intersections AS data_flow
    KEY (edge_id)
    SOURCE KEY (node_id_src) REFERENCES column_node (node_id)
    DESTINATION KEY (node_id) REFERENCES column_node (node_id)
    LABEL data_flow
    PROPERTIES (
      source_schema, source_table, source_column,
      target_schema, target_table, target_column,
      valid_from, valid_to
    )
)
OPTIONS (TRUSTED MODE, DISALLOW MIXED PROPERTY TYPES)
```


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# Query the LINEAGE Graph

```
SELECT DISTINCT
  n.schema_name||'.'||n.table_name||'.'||n.column_name AS id,
  n.schema_name, n.table_name, n.column_name
FROM
  GRAPH_TABLE (via_lineage_graph            -- columns upstream of the table
    MATCH
      (n IS column_node)
      -[e IS data_flow WHERE e.valid_from <= DATE '2026-08-24' AND (e.valid_to IS NULL OR e.valid_to > DATE '2026-08-24')]
      ->{1,2}(tgt IS column_node WHERE tgt.schema_name='HISTORY_NFL' AND tgt.table_name='HISTORY_ALL_POSITIONS_BY_WEEK')
    COLUMNS (n.schema_name, n.table_name, n.column_name))
UNION
SELECT DISTINCT
  n.schema_name || '.' || n.table_name || '.' || n.column_name,
  n.schema_name, n.table_name, n.column_name
FROM
  GRAPH_TABLE (via_lineage_graph            -- columns downstream of the table
    MATCH
      (src IS column_node WHERE src.schema_name='HISTORY_NFL' AND src.table_name='HISTORY_ALL_POSITIONS_BY_WEEK')
      -[e IS data_flow WHERE e.valid_from <= DATE '2026-08-24' AND (e.valid_to IS NULL OR e.valid_to > DATE '2026-08-24')]
      ->{1,2}(n IS column_node)
    COLUMNS (n.schema_name, n.table_name, n.column_name))
UNION
SELECT                                      -- the table's own columns
  schema_name||'.'||table_name||'.'||column_name,
  schema_name, table_name, column_name
FROM mv_via_nodes_columns
WHERE schema_name='HISTORY_NFL' AND table_name='HISTORY_ALL_POSITIONS_BY_WEEK';
```


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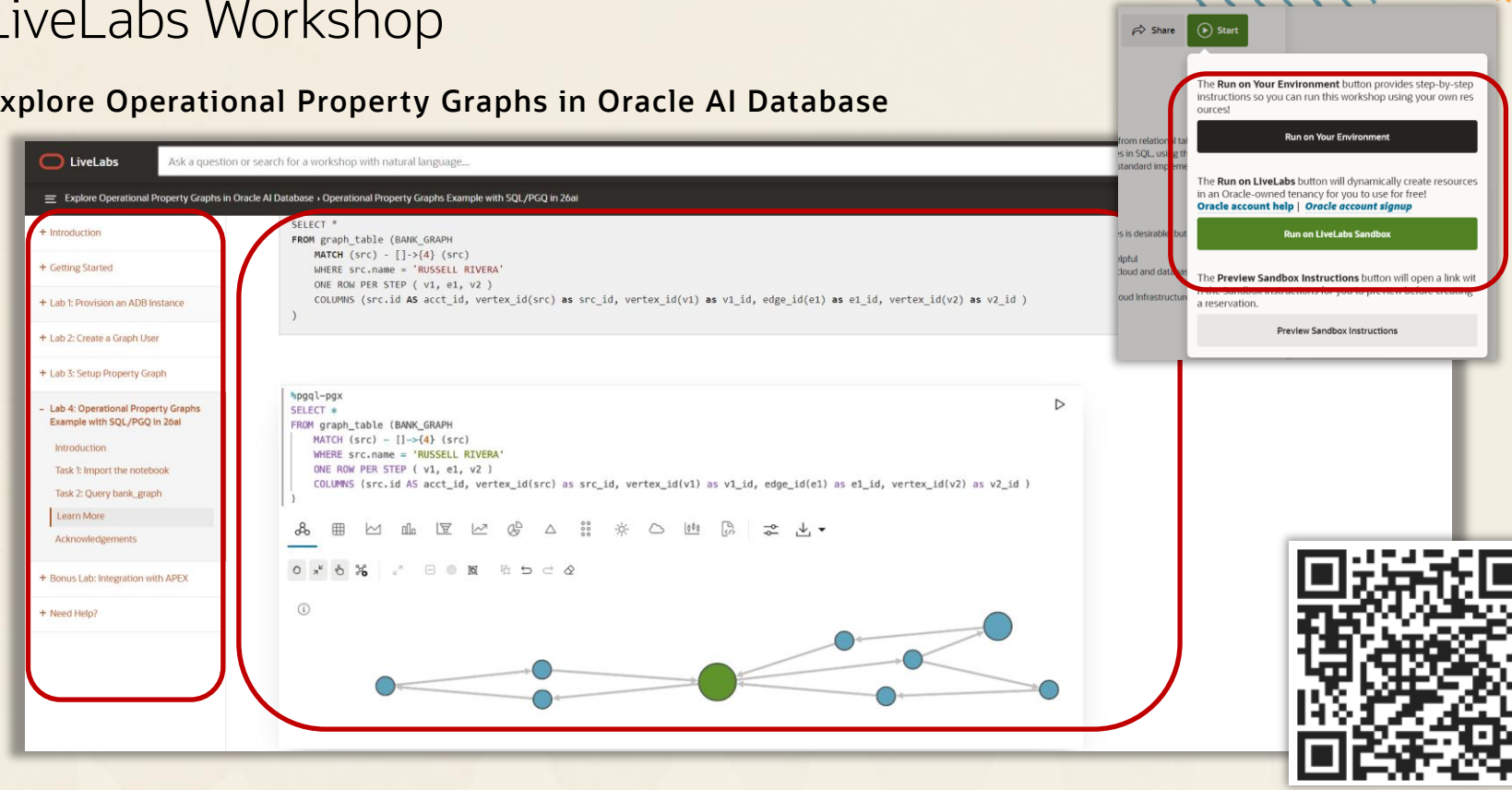
# Getting started

## Try it out yourself


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# LiveLabs Workshop

## Explore Operational Property Graphs in Oracle AI Database





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# Summary


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- Graphs offer a different view of your data, regardless of your business and use case
- With Oracle 26ai, use your existing tools and APIs to work with graphs
- Oracle AI Database is the Graph database for the enterprise
- Developing your Graph-powered applications made simple


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# More information


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# Useful Resources (1/2)

## General information

- www.oracle.com/database/graph/
- Graph Size Estimator
  www.oracle.com/webfolder/assets/graph-size-estimator/index.html
- PGQL Spec 2.1 pgql-lang.org/spec/2.1/

## Documentation Release 26.3

- Oracle Property Graph Developer´s Guide
  docs.oracle.com/en/database/oracle/property-graph/26.3/index.html
- Property Graph Visualization Developer's Guide and Reference
  docs.oracle.com/en/database/oracle/property-graph/26.3/pgvtr/graph-visualization-library-reference.html
- Python API
  docs.oracle.com/en/database/oracle/property-graph/26.3/pypgx/pypgx/api.html
- Java API
  docs.oracle.com/en/database/oracle/property-graph/26.3/spgjv/index.html
- Graph Server REST API
  docs.oracle.com/en/database/oracle/property-graph/26.3/spgdg/graph-server-rest-api-version-2.html

## GitHub

- Machine Learning Workshop with Oracle Property Graph
  github.com/operard/mlgraph
- github.com/oracle-samples/oracle-graph

## Oracle Cloud MarketPlace

- Graph Server and Client – Image
  cloud.oracle.com/marketplace/application/74883066

## YouTube

- youtube.com/@oraclespatialandoraclegraph
- Analytics and Data Techcasts: youtube.com/@analyticsanddatatechcasts90

## Blogs

- Oracle Database Insider – Graph: blogs.oracle.com/database/category/db-graph
- Medium: medium.com/tag/oracle-graph
- Database & Cloud Technology: blogs.oracle.com/coretec/

## LinkedIn

- Oracle Spatial and Oracle Graph group: www.linkedin.com/groups/1848520/


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# Useful Resources (2/2)

- Oracle SQL Developer for VSCode
  marketplace.visualstudio.com/items?itemName=Oracle.sql-developer
- Oracle SQL Developer Graph Visualization for VSCode
  marketplace.visualstudio.com/items?itemName=Oracle.sql-developer-graph-visualization
- Graph Server and Clients downloads
  oracle.com/database/technologies/spatialandgraph/property-graph-features/graph-server-and-client/graph-server-and-client-downloads.html
- Oracle Graph Quickstart Container image
  container-registry.oracle.com/ords/ocr/ba/database/graph-quickstart
- APEX
  - Sample Graph App
    github.com/oracle/apex/tree/26.1/sample-apps/sample-graph-visualizations
  - Graph Visualization Plugin
    github.com/oracle/apex/tree/26.1/plugins/region/graph-visualization
- Get Started with APEX Graph Visualization Plug-in
  docs.oracle.com/en/cloud/paas/autonomous-database/csgru/get-started-apex-graph-visualization-plug.html
- Oracle Graph samples on GitHub
  github.com/oracle-samples/oracle-graph
- Linkurious Ogma integration with Oracle SQL Property Graphs (including a Node.js sample app)
  linkurious.github.io/ogma-oracle-parser/getting-started.html


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# Thank you

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